

Harshwardhan Pandey

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PROFILE

Smart India Hackathon National Finalist (top team from 1,000+ entries) and patent-filing CS engineer who builds AI systems that ship to real users — not demos. At AgriCart, I identified that the LLM integration had no fault-tolerance layer before any production traffic hit it; I designed and built the retry, rate-limiting, and circuit-breaker infrastructure before go-live, entirely on my own initiative. I independently architected a multi-agent RAG platform serving real agricultural advisory queries for smallholder farmers, fine-tuned open-source LLMs using LoRA on 8 GB of domain data, and filed a patent combining generative AI with causal reasoning and blockchain provenance. I move fast, make architectural decisions without hand-holding, and take ownership of outcomes — not just tasks.

WORK EXPERIENCE

Full Stack & AI Developer | AgriCart Farmtech Pvt. Ltd. | *Jul 2024 – Jan 2025*

- **Reduced LLM hallucination rate by 35%** (validated via A/B testing) by independently fine-tuning open-source LLMs on 8 GB of agricultural domain data using LoRA adapters and local model tuning on a GPU instance.
- **Cut average API response latency ~40%** under 50 concurrent users by identifying that synchronous LLM calls were the bottleneck — then redesigning the Spring Boot microservice endpoints with streaming output, exponential-backoff retry logic, and token-bucket rate limiting.
- **Designed and shipped a production RAG pipeline with multi-agent LangChain orchestration** over 12,000 curated ICAR instruction-output pairs — reducing downstream advisory errors ~60% vs. the baseline retrieval-only approach.
- **Compressed release cycles from 5 days to under 1 day** by containerising 3 microservices with Docker and Kubernetes and building Azure DevOps CI/CD pipelines — reducing manual deployment steps from 8 to 1.
- **Eliminated ~60% of downstream data errors before model training** by independently building tokenisation, validation, and embedding pipelines across 8 GB of data from 12 source systems.

Blockchain & Data Analyst | Deloitte — Capstone Programme (Student Finalist) | *Apr 2024 – Jul 2024*

- **Reduced manual reconciliation queries by 40%** across 6 supply chain nodes by analysing distributed ledger transaction data, identifying traceability gaps, and producing structured audit trail reports.
 - **Built and shipped a Vue.js 3 interactive dashboard from scratch** for 3 cross-functional stakeholder teams — replacing static monthly reports with filterable, real-time blockchain audit trail visualisations and cutting sign-off time by one full review cycle.
 - Selected as a top-performing finalist for the programme; independently delivered the end-to-end blockchain supply chain audit system within the 4-month cohort.
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PROJECTS

AgroTrack | *Python | Spring Boot | Vue.js 3 | LangChain | PostgreSQL | Docker | Kubernetes*

- Built end-to-end for smallholder farmers who previously had no access to expert agricultural advisory: Spring Boot REST microservice backend, Vue.js 3 frontend with AG Grid advisory tables, and a multi-agent LangChain RAG pipeline over 12,000 ICAR instruction-output pairs — replacing a ₹2,000/visit agronomist consultation with a sub-2s query response at 50 concurrent users.
- Chose chunked streaming over full-response delivery after profiling revealed 2.1s median latency on batch mode — cutting perceived response time to under 400ms at p50; added circuit-breaker retry logic

and Docker/Kubernetes orchestration, all designed and implemented independently. github.com/Harsh-debug04/agrotrack

Sentiment Intelligence Dashboard | *Python | Spring Boot | React.js | BERT | PostgreSQL | Docker*

- Designed and shipped a full-stack NLP application: chose fine-tuned BERT over a zero-shot classifier after benchmarking showed 23% accuracy gap on domain-specific review language — then trained on 50,000 reviews achieving 91% accuracy across 3 sentiment classes, with a Spring Boot REST API backend and real-time React.js chart frontend.
- Wrote JUnit tests covering 80%+ of API endpoints; containerised with Docker; automated CI build triggered on every push. github.com/Harsh-debug04/sentiment-dashboard

Power Grid Failure Predictor | *Java | Spring Boot | Vue.js 3 | AG Grid | PostgreSQL | Azure DevOps*

- Built a Spring Boot microservice exposing a gradient boosting model (82% accuracy) trained on 3 years of Delhi substation data; consumed by a Vue.js 3 dashboard with AG Grid failure-event tables and drill-down filtering.
- Configured an Azure DevOps CI/CD pipeline reducing manual deployment steps from 8 to 1.

Interpretable Disease Diagnosis | *Python | Spring Boot | React.js | SHAP | LIME | PostgreSQL*

- Analysed 12,000 patient records; applied SHAP and LIME to generate clinically interpretable AI reports delivered through a role-based React.js interface.

PATENT

AI + Causal Reasoning + Blockchain Provenance for Autonomous R&D Pipeline | *Application No. 202611004448 | Filed Jan 2026*

- Invented a tri-core autonomous R&D pipeline: generative AI with domain-specific RAG synthesises novel concepts, a causal reasoning layer eliminates hallucinations via counterfactual simulation (replacing correlation with causation), and a blockchain provenance layer mints each validated idea's cryptographic hash onto an immutable ledger.
- Addresses three production AI problems simultaneously: LLM hallucination in technical domains, R&D vetting bottleneck (weeks-to-months compressed to seconds), and AI intellectual property attribution.

TECHNICAL SKILLS

Backend: Java, Kotlin, Spring Boot, Python, Node.js, REST APIs, Streaming APIs, Microservices

Frontend: Vue.js 3, React.js, HTML5, CSS3, JavaScript ES6+, AG Grid

AI / ML: LLM fine-tuning (LoRA), local model tuning, multi-agent systems, RAG pipelines, LangChain, BERT, SHAP, LIME, Scikit-learn

DevOps: Docker, Kubernetes, Azure DevOps, CI/CD pipelines, Git

Databases: PostgreSQL, MySQL, MongoDB

Testing: JUnit, unit & integration testing, A/B testing, API debugging

EDUCATION

B.Tech — Computer Science Engineering | *Manipal University Jaipur | Expected 2026*

LEADERSHIP

- **Director, Unreal Engine Club (Epic Games Official Partner)** — Hosted workshops; coordinated cross-club events; secured sponsorships from Red Bull and Epic Games.
- **Organising Committee, Annual University Carnival** — Core logistics and programming across parallel tracks.p